RIGHT now
↳ instant speed → derivatives.

Your salary $ increases every year.
Business profit changes daily.
stock prices.
} → Rate of change
↓
derivatives.

② → constant

$y = x^2 \rightarrow y = 4$

$\frac{dy}{dx} = 2x \qquad y = 4.$

$x = 2$

$f(x) = x^2$
$x = 3$
$y = 9$
$y = 6$

$\frac{dy}{dx} = 10$

$y = x^2$

x	y
1	1
2	4
3	9
4	16

derivative measure.
x = time
y = distance
$\frac{dy}{dx} = 10$

building accuracy of
ag ai model.

image → Apple
model → 100%
probi. ↓ 0.8 ← ① $\boxed{0.2}$

Maxima & Minima

↓ highest value
↓ lowest value

Situation	Goal
Profit	Maximum
Marks	Maximum
Error	Minimum
fuel consumption	minimum.
loss	minimum

smaller loss = better predictions.

$f(x) = x^2$

x	$f(x)$
-2	4
-1	1
0	0
1	1
2	4
3	9

at $x = 0$
$f(x)$ → smallest
minimum.

$f(x) = x^2$
$f'(x) = 2x$
$f''(x) = 2$
$f'''(x) = 0$

$2x = 0$
$x = 0$

Shifted.

$f(x) = (x-3)^2 + 2$

$f'(x) = 2(x-3)$

$2(x-3) = 0$

$\Rightarrow x = 3$

minimum point $(3, 2)$

$f(3) = (3-3)^2 + 2$
$f(3) = 2$

$\boxed{-ve.}$ maxima.

$f'(x) = 2(x-3) = 2x - 6$

$f''(x) = 2 > 0$

$f(x) = -x^2 + 4$
$f'(x) = -2x$
$f''(x) = -2 < 0$ ← maximum
$-2x = 0$
$x = 0$

$(0, 4)$

Optimization intuition

	Optimization Goal
Uber	shortest route
Youtube	best recommendation
Amazon	recommendation.
AI model	lowest loss

AI Goal → Minimum Loss

Loss measure → How WRONG the model is

$f(x, y) = x^2 + y^2$
dependency
x = weight 1
y = weight 2

$\frac{\partial f}{\partial x} = 2x$

$\frac{\partial f}{\partial y} = 2y$

Gradient

AI → high loss ∝
↓
low loss
↓
Gradient descent.
moving opposite to gradient

Temperature.

x = time
y = temperature

at 10:00 AM → temperature rise quickly
Gradient tells → temperature is increasing very fast
If gradient is large +ve, tem ↑ quickly.

Water tank. → x = time
y = water amount. → How fast tank is filling.

Numerical differentiation.

↳ approximation.

$$\frac{f(x+h) - f(x)}{h}$$

$\boxed{h} \rightarrow 0.0001$

$f(x) = x^2$

$x = 2$

$(2.0001)^2$

$h = 0.0001$

$$\frac{f(x+h) - f(x)}{h} = \frac{f(2.0001) - f(2)}{0.0001}$$

$f(2) = 4$

$f(2.0001) = 4.0004001$

$x^2 \rightarrow 2x$
at $x = 2 \rightarrow 4$

Jacobian matrix

many inputs → many outputs.
multiple inputs + multiple outputs.

$f(x, y) = \begin{cases} x^2 + y \\ 2x + y^3 \end{cases}$

$2 \rightarrow x, y$
$2 \rightarrow f_1, f_2$

$f_1(x, y) = x^2 + y$
$f_2(x, y) = 2x + y^3$

every output derivative with every input.

$$\begin{array}{c|cc} & x & y \\ \hline f_1 & \partial f_1/\partial x & \partial f_1/\partial y \\ f_2 & \partial f_2/\partial x & \partial f_2/\partial y \end{array}$$

Jacobian matrix

$f_1 = x^2 + y \rightarrow 2x + 1$
$f_2 = 2x + y^3 \rightarrow 2 + 3y^2$

$$\Rightarrow \begin{bmatrix} 2x & 1 \\ 2 & 3y^2 \end{bmatrix}$$

outputs ← $m \times n$ → inputs

Hessian matrix.

first derivative table
second derivative table.
- curvature
- maxima/minima
- optimization
- neural network training.

$f(x, y) = x^2 + 3xy + y^2$

↳ $\frac{\partial f}{\partial x} = 2x + 3y \rightarrow 2, 3$

$\frac{\partial f}{\partial y} = 3x + 2y \rightarrow 3, 2$

~~3y~~ $3x + 2y$

$$\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \qquad \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial^2 y} \end{bmatrix}$$

s → ~~speed~~ distance

$\frac{ds}{dt}$ → speed

$\frac{d^2 s}{dt^2}$ = acceleration.

→ multi-direction acceleration map.

$f(x, y) = x^2 + xy + y^2$

$\frac{\partial f}{\partial x} = 2x + y \rightarrow \frac{\partial^2 f}{\partial x^2} = 2$

$\frac{\partial f}{\partial y} = x + 2y \rightarrow \frac{\partial^2 f}{\partial y^2} = 2$

$\frac{\partial f}{\partial x} = 2x + y \rightarrow \frac{\partial f}{\partial x \partial y} = 1$

$\frac{\partial f}{\partial y} = x + 2y \rightarrow \frac{\partial f}{\partial y \partial x} = 1$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Term	Meaning
first derivative	Slope
second derivative	curvature
gradient	all first derivatives
Jacobian	first derivative table
Hessian	second derivative table

Scipy optimization

↳ python to find the best value of something.
scipy.optimize

$f(x) = x^2 + 4x + 4$
$f'(x) = 2x + 4$
$f''(x) = 2 > 0$

$2x + 4 = 0$
$2x = -4$
$x = -2$

$f(x) = 2x^2 + 5x + 3$
$f'(x) = 4x + 5$
$f''(x) = 4 > 0$

There is no maxima

minima.
↓
$4x + 5 = 0$
$x = -\frac{5}{4}$

$f(x) = -2x^2$
$f'(x) = -4x$
$f''(x) = -4 < 0$
↓
maxima.

~~4x~~ = $-4x = 0$
$x = 0$